CHEAT SHEET · JS INTERVIEW #07 @specsycoder

typeof null "object"

A 1995 bug that can never be fixed

null.js

```
1  typeof null; // "object"
2  null === null; // true
```

output "object"

Why it happens

01 Type tags
The first JS engine stored every value with a small type tag.

02 Tag 0 = object
Objects were marked with the tag 0.

03 null = all zeros
null was stored as all zeros, so its tag also read as 0.

04 Never fixed
A fix was proposed, but it broke existing websites, so it was dropped.

Check for null the right way

check.js

```
1  // only null
2  value === null
3  // null OR undefined
4  value == null
5  // a real object
6  value !== null && typeof value === "object"
```

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More typeof surprises

And the right check for each

typeof.js

```
1 typeof []; // "object"
2 typeof NaN; // "number"
3 typeof function () {}; // "function"
4 typeof notDeclared; // "undefined", no error
```

value	typeof says	use instead
null	"object"	value === null
[1, 2, 3]	"object"	Array.isArray(value)
NaN	"number"	Number.isNaN(value)
new Date()	"object"	value instanceof Date

Traps to mention

- `== null` **is a handy shortcut.** It matches both null and undefined, and nothing else.
- **typeof never throws** for undeclared variables, except let/const in their TDZ (ReferenceError).
- **Functions are objects too,** but typeof gives them their own answer: "function".

Your 20-second answer

"typeof null returns 'object'. It is a bug from the first version of JavaScript: values carried a type tag, objects used tag 0, and null was stored as all zeros, so it read as an object. Fixing it would break existing sites, so it stays. To check for null, I compare value === null."